THE UK UNIVERSITY INTEGRATION BEE 2024/2025

TOURNAMENT ROUND

Sponsored by:  Jane Street

BATTLE 1

BATTLE 1

Fight 1

Declare Score Gamble

Problem 1

Problem 1

Problem 1

$$\int_0^\infty \frac{\lfloor x \rfloor}{1 + \lfloor x \rfloor^2 + \lfloor x \rfloor^4} \, \mathrm{d}x$$

Problem 1

Answer

Problem 1

$$\frac{1}{2}$$

Problem 2

Problem 2

Problem 2

$$\int_0^{\frac{\pi}{4}} (4\cos^2(x) - 1)\sqrt{\tan(x)}\,\mathrm{d}x$$

Problem 2

Answer

Problem 2

1

Problem 3

Problem 3

Problem 3

$$\int_0^{\frac{\pi}{2}} \frac{2 - \sqrt{\tan(x)} + \tan(x)}{1 + \tan^{3/2}(x)} \, \mathrm{d}x$$

Problem 3

Answer

Problem 3

$$\frac{\pi}{2}$$

BATTLE 1

Fight 2

Declare Score Gamble

Problem 4

Problem 4

Problem 4

$$\int \frac{x-1}{(x-2)(x-3)} + \frac{x-2}{(x-3)(x-1)} + \frac{x-3}{(x-1)(x-2)}\,\mathrm{d}x$$

Problem 4

Answer

Problem 4

$$\frac{5}{2}\ln|x-3| + \frac{5}{2}\ln|x-1| - 2\ln|x-2| + C$$

Problem 5

Problem 5

Problem 5

$$\int_1^{\infty} \frac{1}{x\left(x^{\ln(2048)}+1\right)}\,\mathrm{d}x$$

Problem 5

Answer

Problem 5

$$\frac{1}{11}$$

Problem 6

Problem 6

Problem 6

$$\int_0^{2\pi} \frac{1}{(1+2^{\sin(x)})(1+2^{\cos(x)})}\,\mathrm{d}x$$

Problem 6

Answer

Problem 6

$$\frac{\pi}{2}$$

BATTLE 1

Fight 3

Declare Score Gamble

Problem 7

Problem 7

Problem 7

$$\int_{-1}^{1} \frac{2025 + e^{x} + e^{-x}}{1 + 2025^{x}} \, \mathrm{d}x$$

Problem 7

Answer

Problem 7

$$2025 + e - \frac{1}{e}$$

Problem 8

Problem 8

Problem 8

$$\int_0^\infty \frac{1}{\sinh(\sqrt{x}) + \operatorname{csch}(\sqrt{x})}\,\mathrm{d}x$$

Problem 8

Answer

Problem 8

π

Problem 9

Problem 9

Problem 9

$$\int_0^1 \sqrt{x + x^3} + \sqrt{x + \sqrt[3]{x}}\, \mathrm{d}x$$

Problem 9

Answer

Problem 9

$$\frac{4\sqrt{2}}{3}$$

BATTLE 2

BATTLE 2

Fight 1

Declare Score Gamble

Problem 1

Problem 1

Problem 1

$$\int_1^e \frac{x \ln(x)}{(1+\ln(x))^3}\,\mathrm{d}x$$

Problem 1

Answer

Problem 1

$$\frac{1}{2}\left(\frac{e^2}{4}-1\right)$$

Problem 2

Problem 2

Problem 2

$$\int_0^\infty \frac{\tan^{-1}(x)}{x((\ln(x))^2+1)}\,\mathrm{d}x$$

Problem 2

Answer

Problem 2

$$\frac{\pi^2}{4}$$

Problem 3

Problem 3

Problem 3

$$\int_1^\infty \frac{1}{x\sqrt{\sqrt[e]{x}-1}}\,\mathrm{d}x$$

Problem 3

Answer

Problem 3

$e\pi$

BATTLE 2

Fight 2

Declare Score Gamble

Problem 4

Problem 4

Problem 4

$$\int \frac{x^2-1}{x^3+x} \ln\left(x^3+\frac{1}{x^3}+3\left(x+\frac{1}{x}\right)\right)\,\mathrm{d}x$$

Problem 4

Answer

Problem 4

$$\frac{3}{2}\left(\ln\left(x+\frac{1}{x}\right)\right)^2+C$$

Problem 5

Problem 5

Problem 5

$$\int_0^1 \frac{\ln(x)(2x-1)}{\sqrt{x-x^2}}\,\mathrm{d}x$$

Problem 5

Answer

Problem 5

$$\pi$$

Problem 6

Problem 6

Problem 6

$$\int_{-2024}^{2025} f_\infty(x)\,\mathrm{d}x$$

$$\text{where } f_n(x) = \begin{cases} \dfrac{2x}{1+x^2}, & n = 0; \\ f_{n-1}(f_{n-1}(x)), & n \geq 1, n \in \mathbb{Z}; \\ \lim\limits_{k\to\infty} f_k(x), & n = \infty \end{cases}$$

Problem 6

Answer

Problem 6

1

BATTLE 2

Fight 3

Declare Score Gamble

Problem 7

Problem 7

Problem 7

$$\int_3^4 \frac{x - \sqrt{7x - x^2}}{2x - 7} \, \mathrm{d}x$$

Problem 7

Answer

Problem 7

$$\frac{1}{2}$$

Problem 8

Problem 8

Problem 8

$$\int_1^{\infty} \frac{1 + 2x \ln(2)}{x\sqrt{x4^x - 1}} \, \mathrm{d}x$$

Problem 8

Answer

Problem 8

$$\frac{\pi}{3}$$

Problem 9

Problem 9

Problem 9

$$\int \sec(x)(\sec(x)+\tan(x))^{2025}\,\mathrm{d}x$$

Problem 9

Answer

Problem 9

$$\frac{(\sec(x) + \tan(x))^{2025}}{2025} + C$$

BATTLE 3

BATTLE 3

Fight 1

Declare Score Gamble

Problem 1

Problem 1

Problem 1

$$\int \frac{\ln(x)+2}{(xe)^{-x}+2}\,\mathrm{d}x$$

Problem 1

Answer

Problem 1

$$\frac{1}{2}\ln(1+2(xe)^x)+C$$

Problem 2

Problem 2

Problem 2

$$\int_0^{\infty} \frac{\tan^{-1}\left(\frac{x}{1+2x^2}\right) - \tan^{-1}\left(\frac{x}{1+12x^2}\right)}{x}\,\mathrm{d}x$$

Problem 2

Answer

Problem 2

$$\frac{\pi}{2}(\ln(3) - \ln(2))$$

Problem 3

Problem 3

Problem 3

$$\int_0^{\frac{\pi}{2}} \sin(x)\sqrt{\sin(2x)}\,\mathrm{d}x$$

Problem 3

Answer

Problem 3

$$\frac{\pi}{4}$$

BATTLE 3

Fight 2

Declare Score Gamble

Problem 4

Problem 4

Problem 4

$$\int_{-\infty}^{\infty} \frac{2^{\lfloor 2^x \rfloor + x}}{3^{2^{\lfloor 2^x \rfloor}} + 1} \, \mathrm{d}x$$

Problem 4

Answer

Problem 4

$$\frac{1}{2\ln(2)}$$

Problem 5

Problem 5

Problem 5

$$\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \sin(x)^{\tan(x)} \left(\sec^2(x)\ln(\sin(x)) + 1\right)\ \mathrm{d}x$$

Problem 5

Answer

Problem 5

$$1 - \frac{\sqrt{2}}{2}$$

Problem 6

Problem 6

Problem 6

$$\int_1^{\sqrt{2}} \frac{\sqrt{x^2 + 2\sqrt{x^2 - 1}}}{x} \, \mathrm{d}x$$

Problem 6

Answer

Problem 6

$$\frac{1}{2}\ln(2) + 1 - \frac{\pi}{4}$$

BATTLE 3

Fight 3

Declare Score Gamble

Problem 7

Problem 7

Problem 7

$$\int_0^\infty \cos\left(x^2 + \frac{1}{x^2}\right)\,\mathrm{d}x$$

Problem 7

Answer

Problem 7

$$\frac{\sqrt{2\pi}}{4}(\cos(2) - \sin(2))$$

Problem 8

Problem 8

Problem 8

$$\int_0^1 x^e\left(e^x + e^{-x}\right) + x^{e-1}\left(e^{x+1} - e^{1-x}\right)\,\mathrm{d}x$$

Problem 8

Answer

Problem 8

$$e - \frac{1}{e}$$

Problem 9

Problem 9

Problem 9

$$\int \frac{1}{\sin(x)(\ln(\sin(x)) - \ln(1+\cos(x)))}\,\mathrm{d}x$$

Problem 9

Answer

Problem 9

$$\ln\left|\ln(\sin(x)) - \ln(1+\cos(x))\right| + C$$

BATTLE 4

BATTLE 4

Fight 1

Declare Score Gamble

Problem 1

Problem 1

Problem 1

$$\int_0^{\infty} \frac{3^{-\lfloor x \rfloor} \tan\left(3^{-\lfloor x \rfloor}\right)}{3 - \tan^2\left(3^{-\lfloor x \rfloor}\right)} \, \mathrm{d}x$$

Problem 1

Answer

Problem 1

$$\frac{1}{8}\left(1-3\cot(3)\right)$$

or

$$\frac{1}{8}\left(1-3\tan\left(\frac{\pi}{2}-3\right)\right)$$

Problem 2

Problem 2

Problem 2

$$\int_0^1 x^3 \sin^2(\ln(x)) \cos^2(\ln(x)) \, \mathrm{d}x$$

Problem 2

Answer

Problem 2

$$\frac{1}{64}$$

Problem 3

Problem 3

Problem 3

$$\lim_{n\to\infty} n \int_0^{\frac{\pi}{4}} \tan^n(x)\, \mathrm{d}x$$

Problem 3

Answer

Problem 3

$$\frac{1}{2}$$

BATTLE 4

Fight 2

Declare Score Gamble

Problem 4

Problem 4

Problem 4

$$\int_0^{\frac{\pi}{2^n}} \frac{\sin\left(\left(2^m + 2025\left(2^n\right)\right)x\right)}{\sin\left(2^m x\right)} \prod_{k=1}^{m-n} \cos\left(2^{m-k}x\right)\,\mathrm{d}x$$

for $m, n \in \mathbb{Z}$, $m > n$

Problem 4

Answer

Problem 4

$$\frac{\pi}{2^{m-2n}}$$

Problem 5

Problem 5

Problem 5

$$\int \frac{e^{W(x)}}{x^2}\,\mathrm{d}x$$

where W is the Lambert W_0 function (i.e. the inverse function to $x = we^w$ on the principal branch)

Note: Answers in terms of $W(x)$ are acceptable

Problem 5

Answer

Problem 5

$$\ln|W(x)| - \frac{1}{W(x)} + C$$

Problem 6

Problem 6

Problem 6

$$\int_{-\infty}^{\infty} \frac{\sin^2(x)}{45x^2 + x^2 \tan^{-1}(x) + x^2\sqrt{2025 + (\tan^{-1}(x))^2}}\,\mathrm{d}x$$

Problem 6

Answer

Problem 6

$$\frac{\pi}{90}$$

BATTLE 4

Fight 3

Declare Score Gamble

Problem 7

Problem 7

Problem 7

$$\int_0^{\frac{\pi}{4}} \frac{1}{\sqrt[4]{2}+\sqrt{1+\tan(x)}}\,\mathrm{d}x$$

Problem 7

Answer

Problem 7

$$\frac{\pi}{8\sqrt[4]{2}}$$

Problem 8

Problem 8

Problem 8

$$\int_0^\infty \frac{\lfloor x \rfloor !(2\lfloor x \rfloor)!!}{(2\lceil x \rceil - 1)!}\,\mathrm{d}x$$

Problem 8

Answer

Problem 8

$$\frac{\pi}{2}$$

Problem 9

Problem 9

Problem 9

$$\int_1^3 \frac{\ln(x)}{x^2+3}\,\mathrm{d}x$$

Problem 9

Answer

Problem 9

$$\frac{\pi \ln(3)}{12\sqrt{3}}$$